Ex.No: 11

Formulae:

i. To compute y_in for an activation function:

$$y_{in} = b + \sum_{i}^{n} x_i \, w_{ij}$$

ii. Sigmoid activation function:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

iii. Mean Squared Error (MSE):

$$\underbrace{\sum_{i=1}^{n} \left(\overbrace{Y_i}^{\text{real value}} - \overbrace{\hat{Y}_i}^{\text{predicted value}} \right)^2}_{\text{sum of the errors of all samples}}$$

number of samples

Note: In our simulation, we used the same bias for every node; however, in practice, biases are often adjusted individually to allow for more flexibility in learning complex patterns.

The ANN architecture consists of input, hidden, and output layers, while the hidden layers perform weighted computations and apply activation functions.

Observations (Simulation Screenshots):

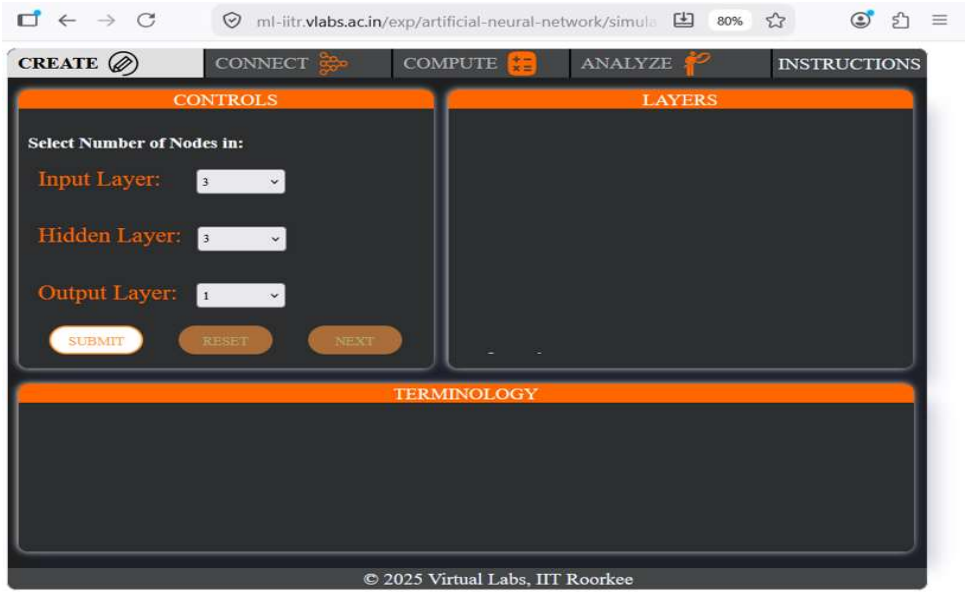


Fig 1: Selecting the number of nodes for the input, hidden, and output layers

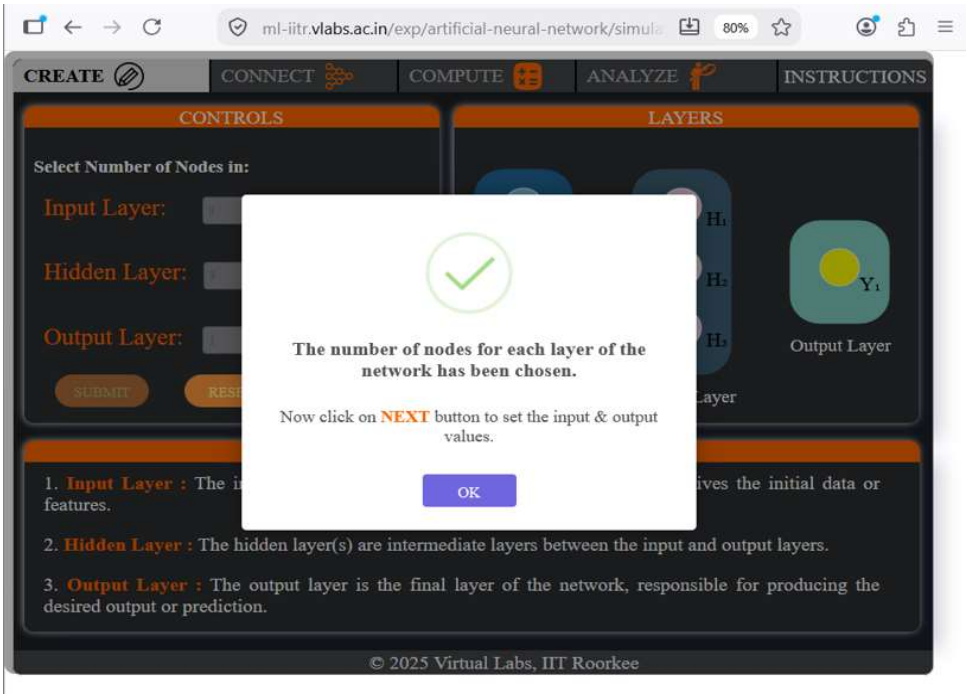


Fig 2: *Confirmation message after the number of nodes is chosen*

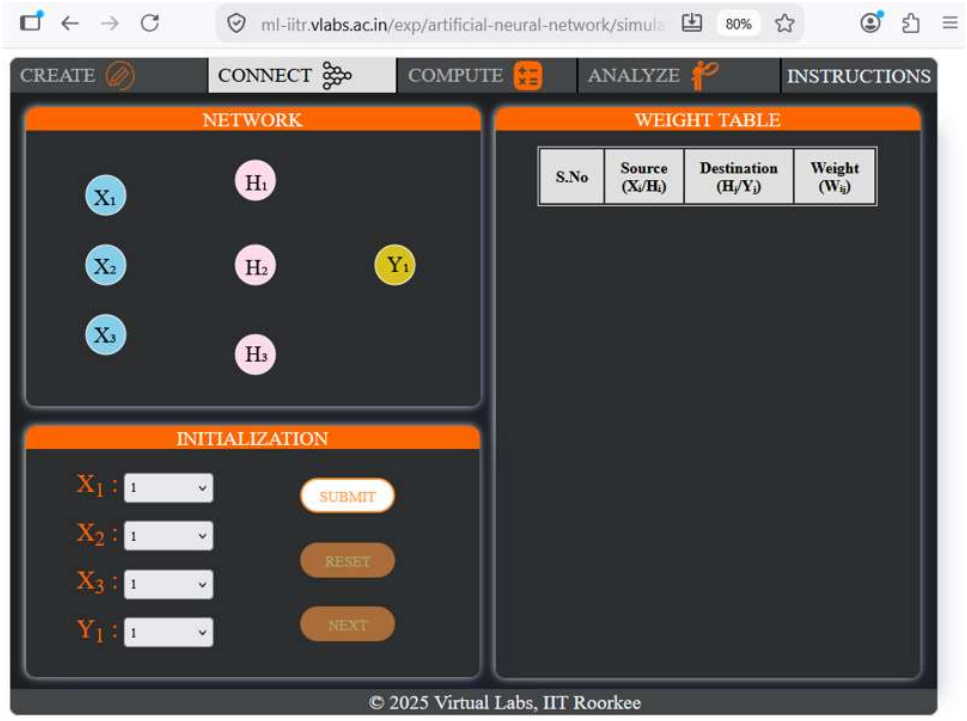


Fig 3: *Network layout with fields to initialize input (Xi) and output (Yi) values*

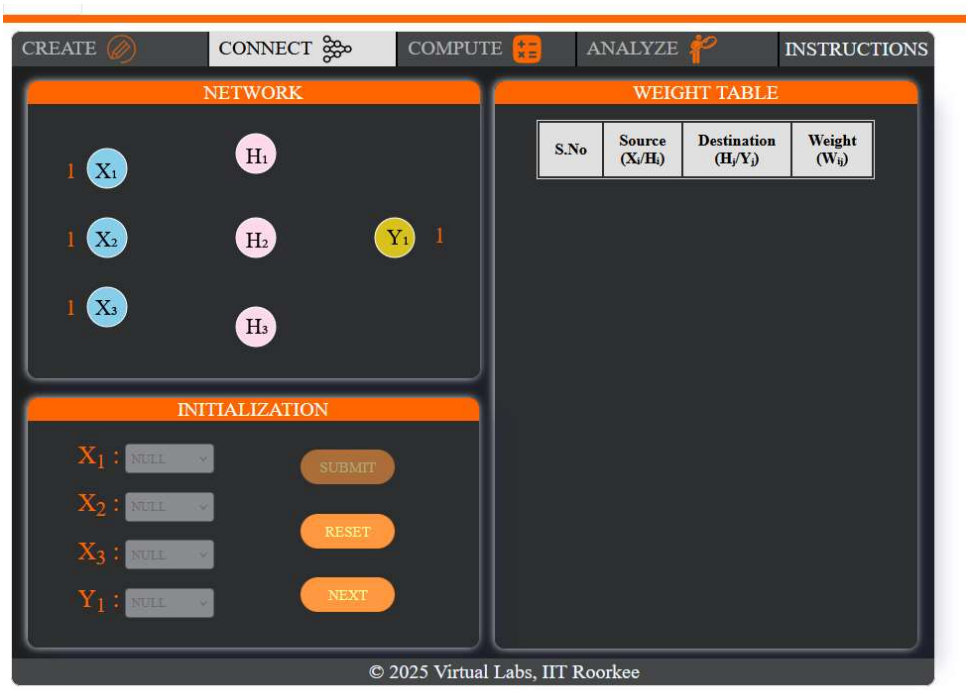


Fig 4: Input and output values submitted (X1 = X2 = X3 = 1, Y1 = 1)

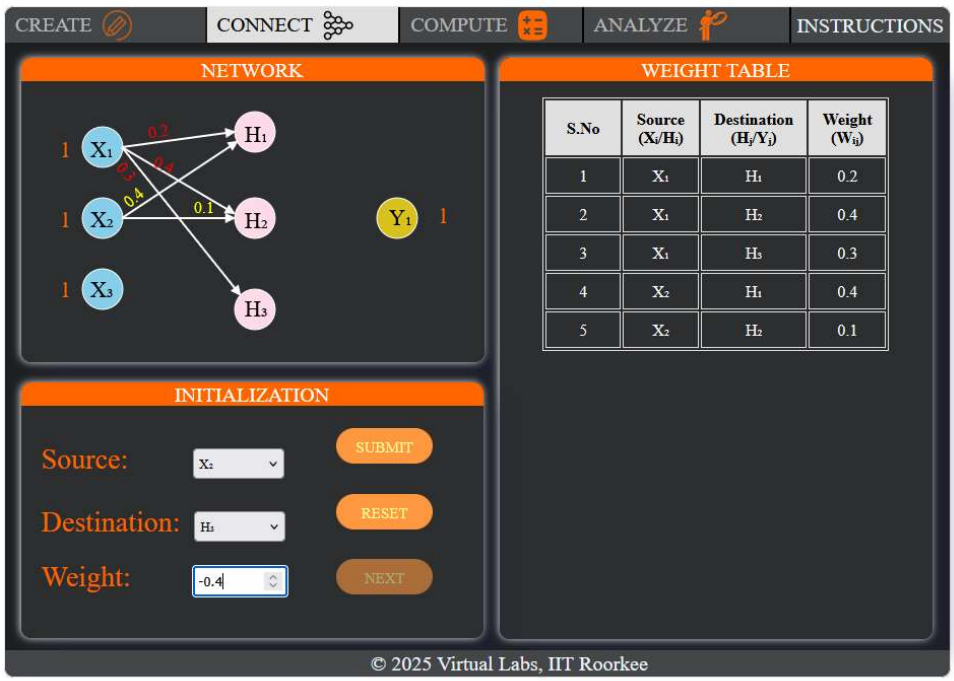


Fig 5: Assigning a weight between an input node and a hidden node (X2 → H2 = -0.4)

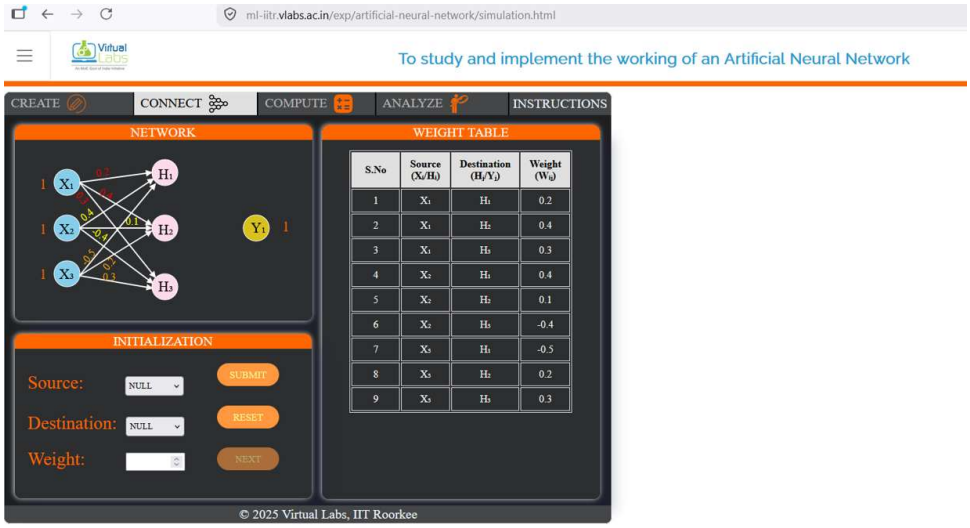


Fig 6: Weight table after all input-hidden layer connections are assigned

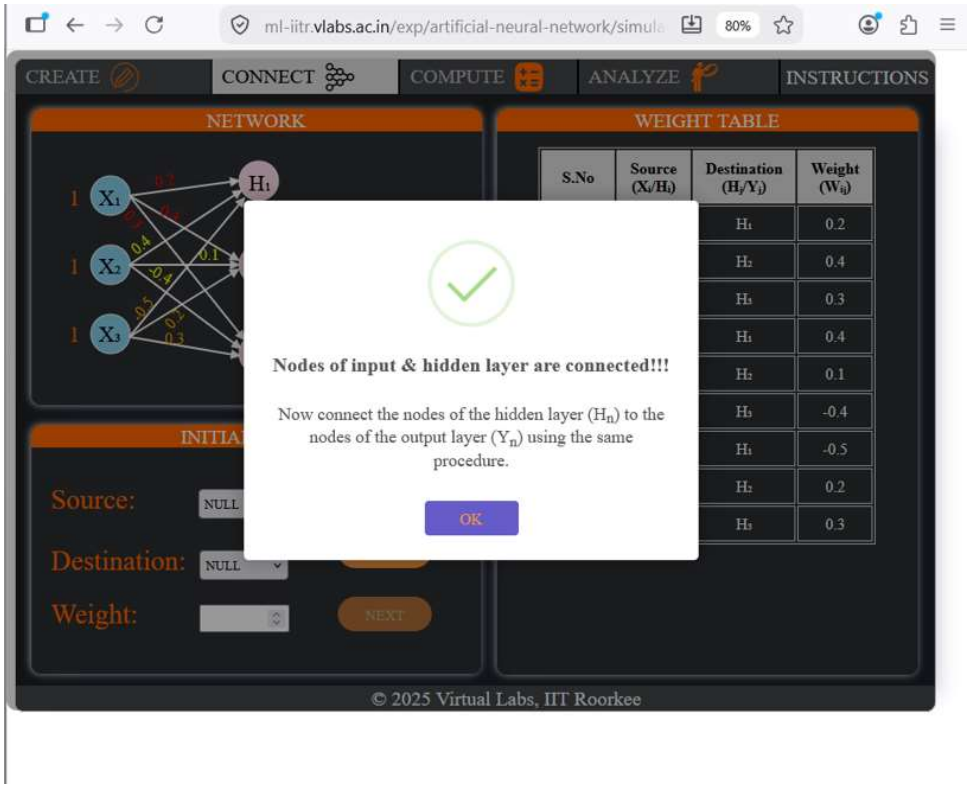
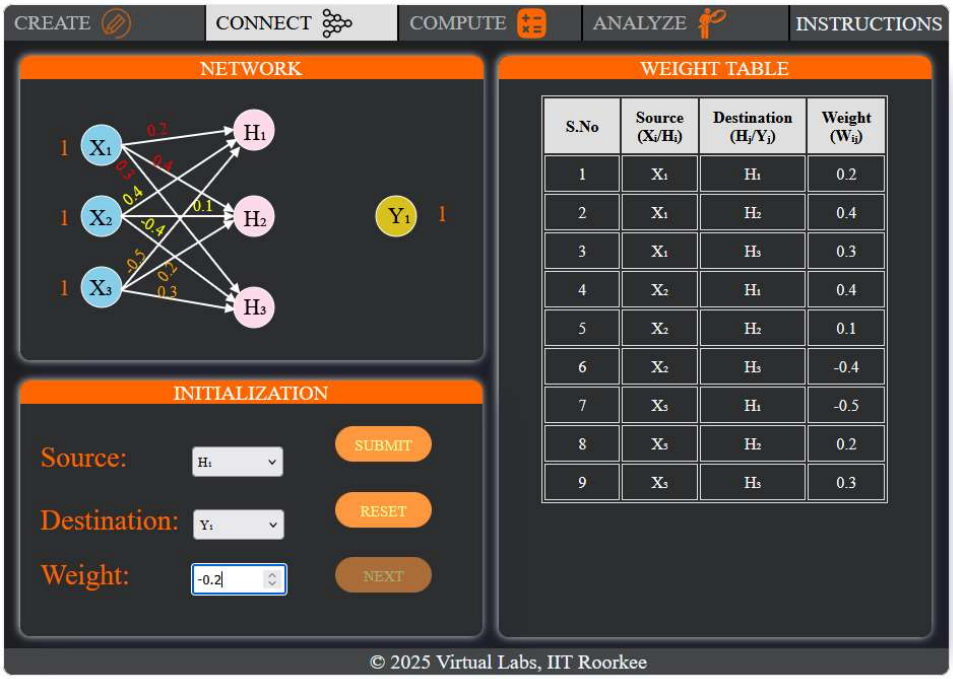
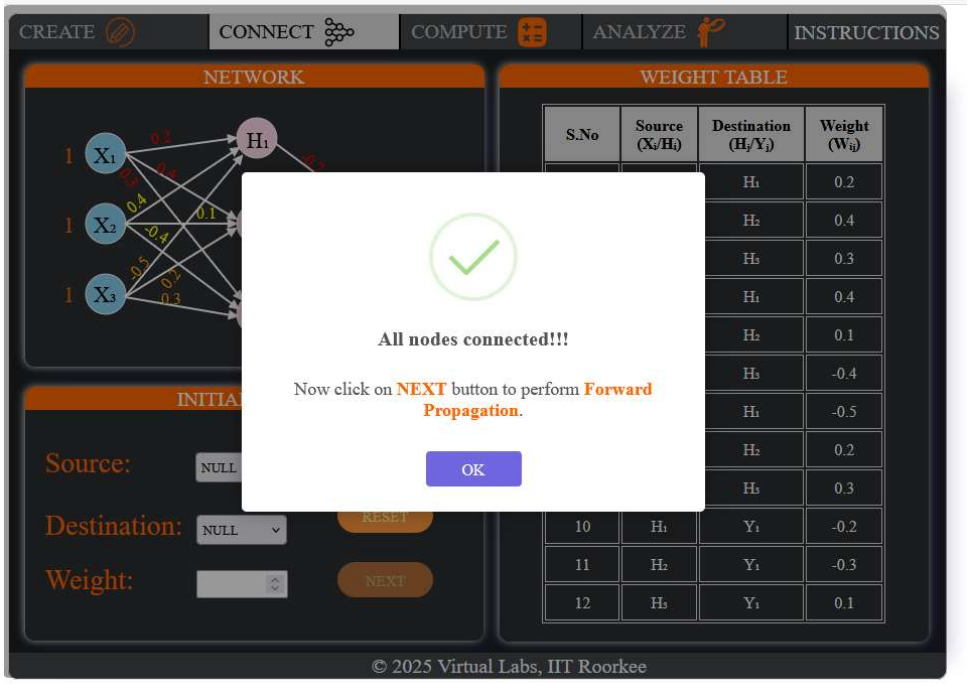


Fig 7: Confirmation – input and hidden layer nodes connected



S.No	Source (X_i/H_i)	Destination (H_j/Y_j)	Weight (W_{ij})
1	X_1	H_1	0.2
2	X_1	H_2	0.4
3	X_1	H_3	0.3
4	X_2	H_1	0.4
5	X_2	H_2	0.1
6	X_2	H_3	-0.4
7	X_3	H_1	-0.5
8	X_3	H_2	0.2
9	X_3	H_3	0.3

Fig 8: *Assigning a weight between a hidden node and the output node (H1 → Y1 = -0.2)*



S.No	Source (X_i/H_i)	Destination (H_j/Y_j)	Weight (W_{ij})
		H_1	0.2
		H_2	0.4
		H_3	0.3
		H_1	0.4
		H_2	0.1
		H_3	-0.4
		H_1	-0.5
		H_2	0.2
		H_3	0.3
10	H_1	Y_1	-0.2
11	H_2	Y_1	-0.3
12	H_3	Y_1	0.1

Fig 9: *Confirmation – all nodes connected, complete weight table (12 connections)*

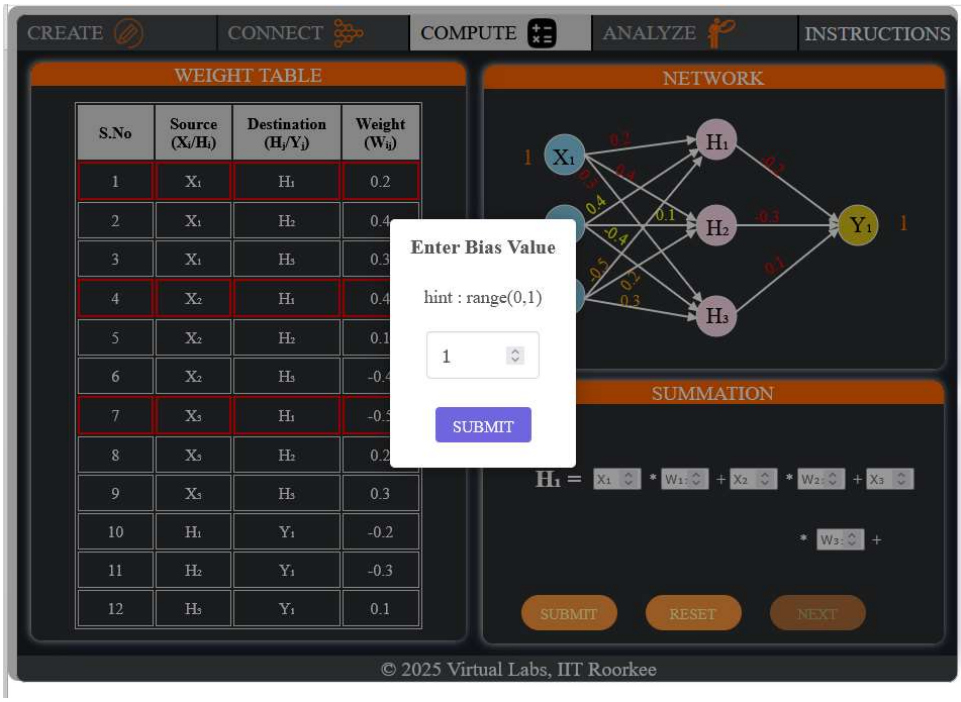


Fig 10: Entering the bias value

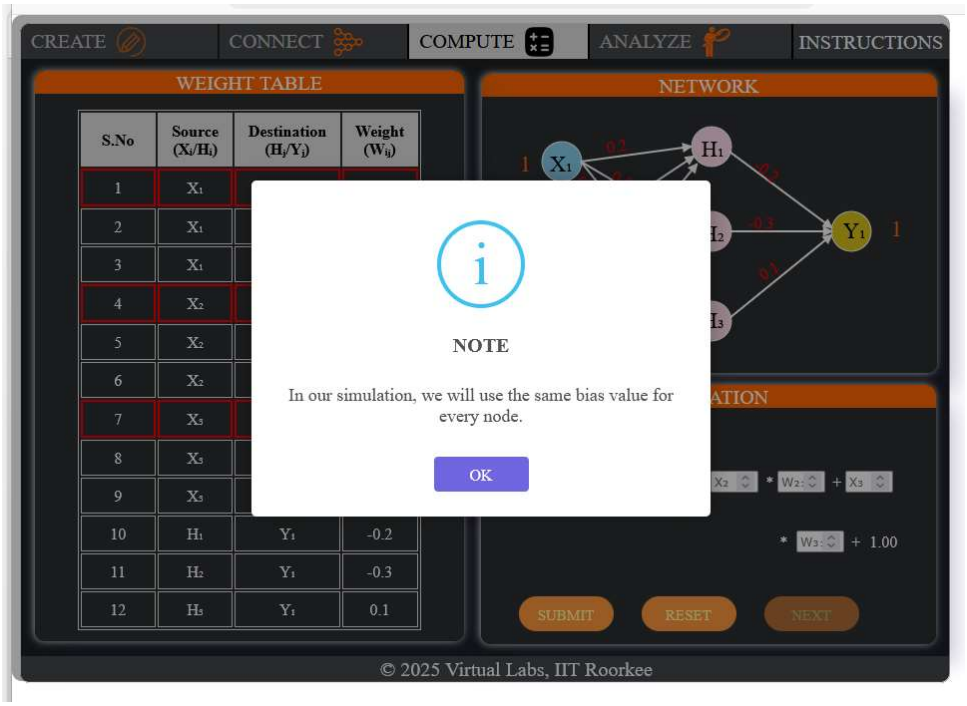
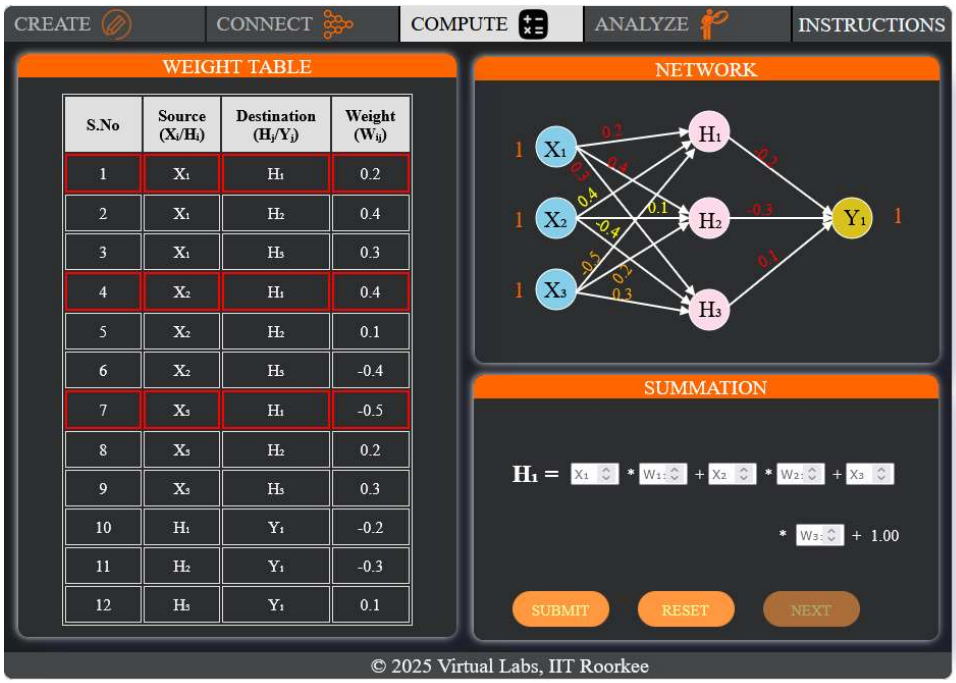


Fig 11: The simulation uses the same bias value for every node



S.No	Source (X_i/H_i)	Destination (H_j/Y_j)	Weight (W_{ij})
1	X_1	H_1	0.2
2	X_1	H_2	0.4
3	X_1	H_3	0.3
4	X_2	H_1	0.4
5	X_2	H_2	0.1
6	X_2	H_3	-0.4
7	X_3	H_1	-0.5
8	X_3	H_2	0.2
9	X_3	H_3	0.3
10	H_1	Y_1	-0.2
11	H_2	Y_1	-0.3
12	H_3	Y_1	0.1

Fig 12: Summation setup for hidden layer node H1

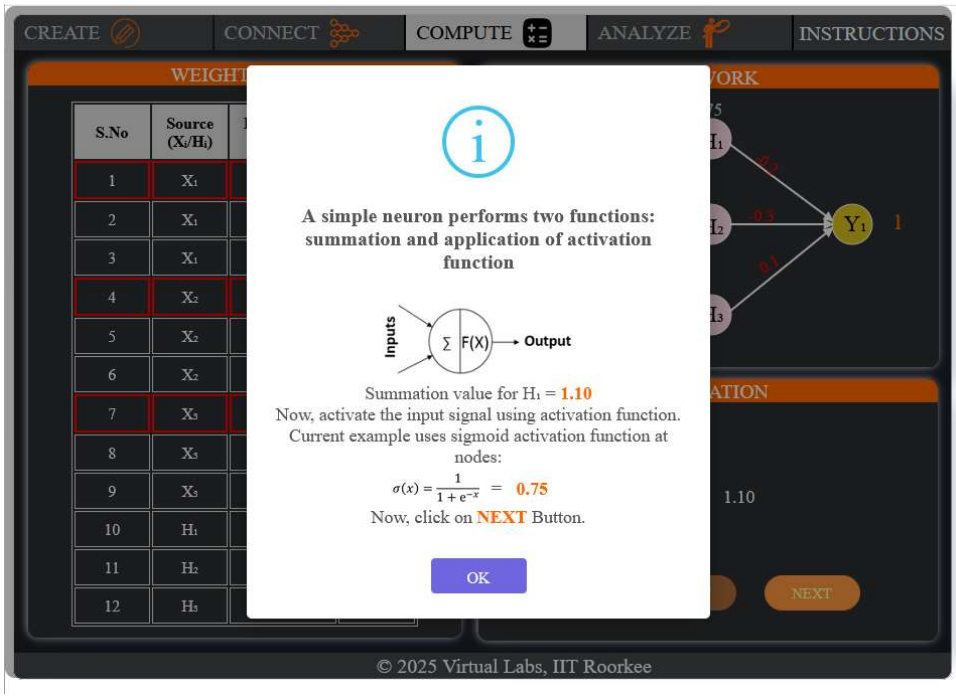


Fig 13: Summation (1.10) and sigmoid activation (0.75) computed for H1

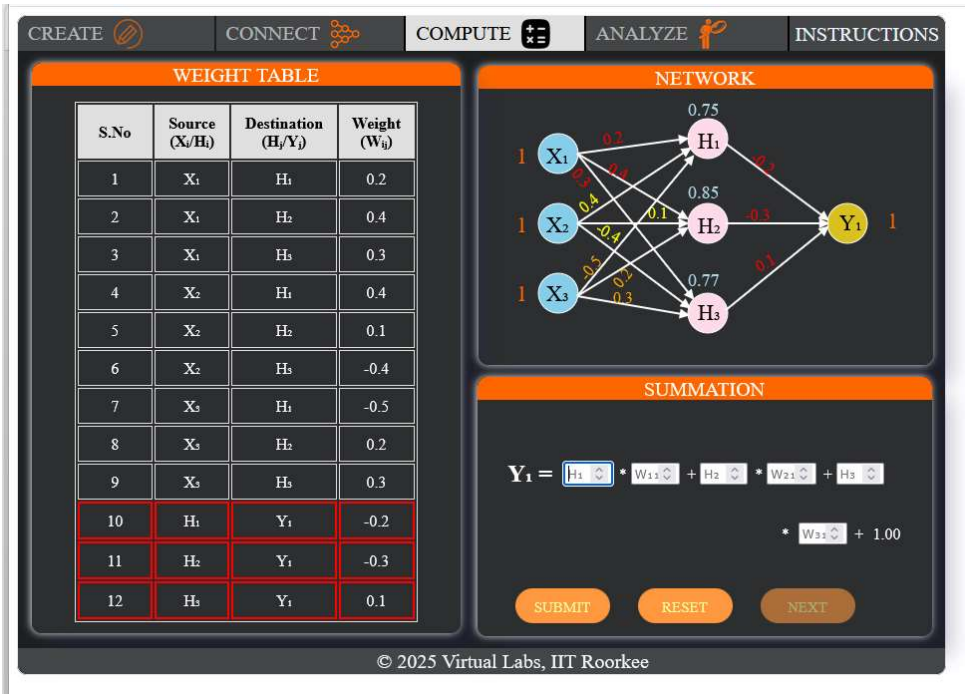


Fig 14: *Hidden layer outputs obtained ; summation setup for output node Y1*

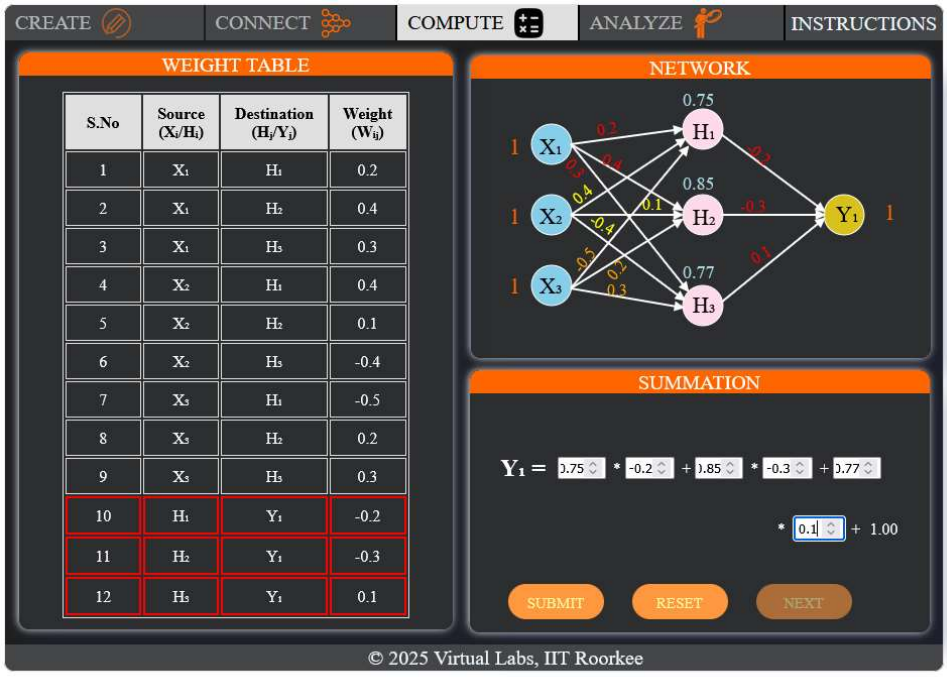


Fig 15: *Output layer summation values entered for Y1*

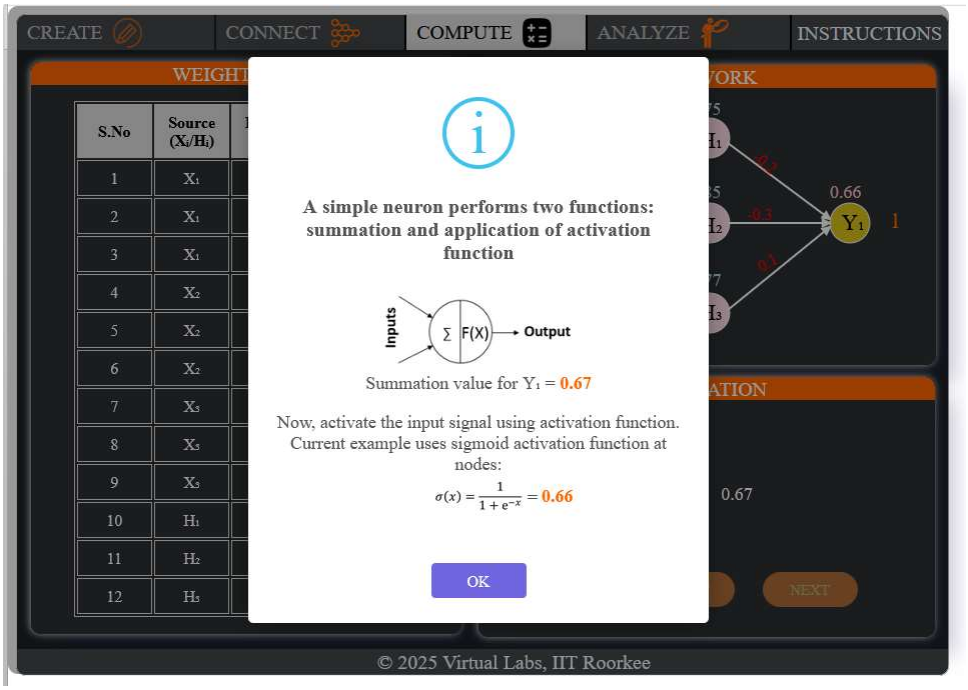


Fig 16: Summation (0.67) and sigmoid activation (0.66) computed for Y1

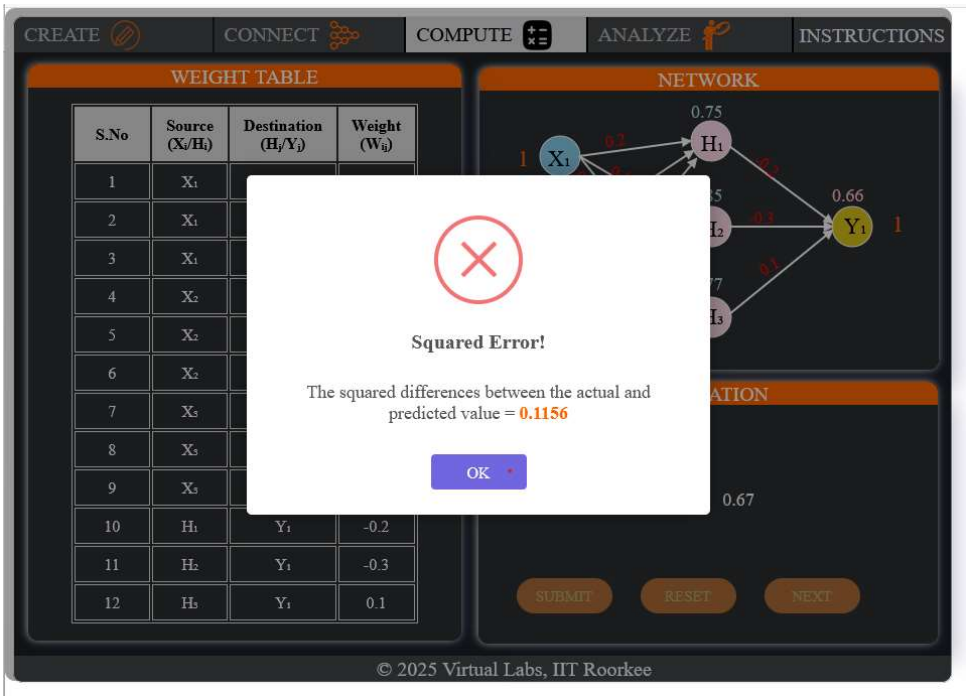


Fig 17: Squared error computed between actual and predicted output (0.1156)

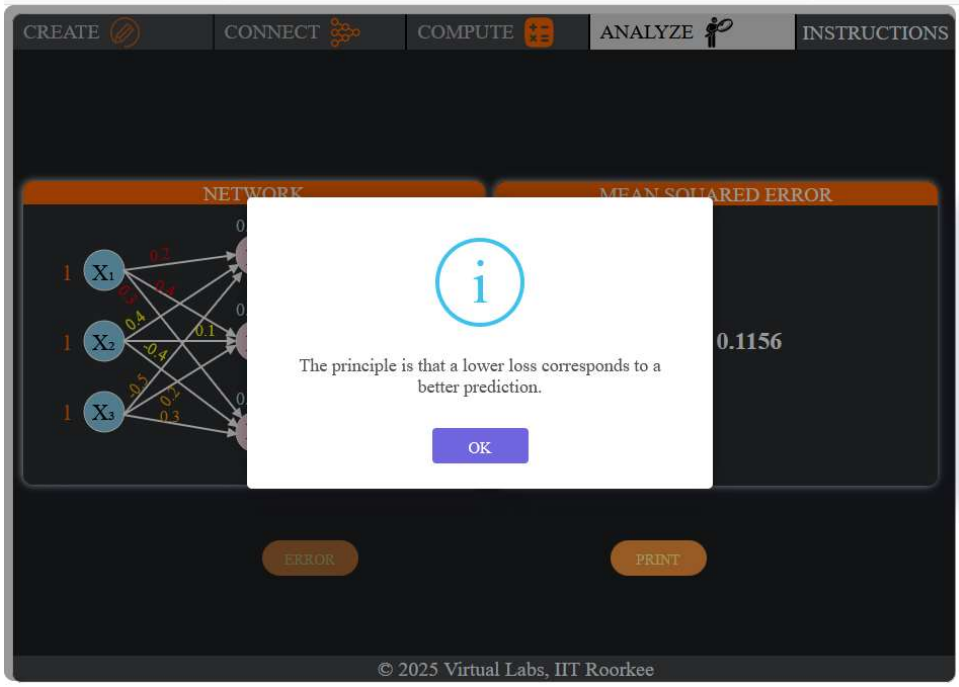


Fig 18: *A lower loss corresponds to a better prediction*

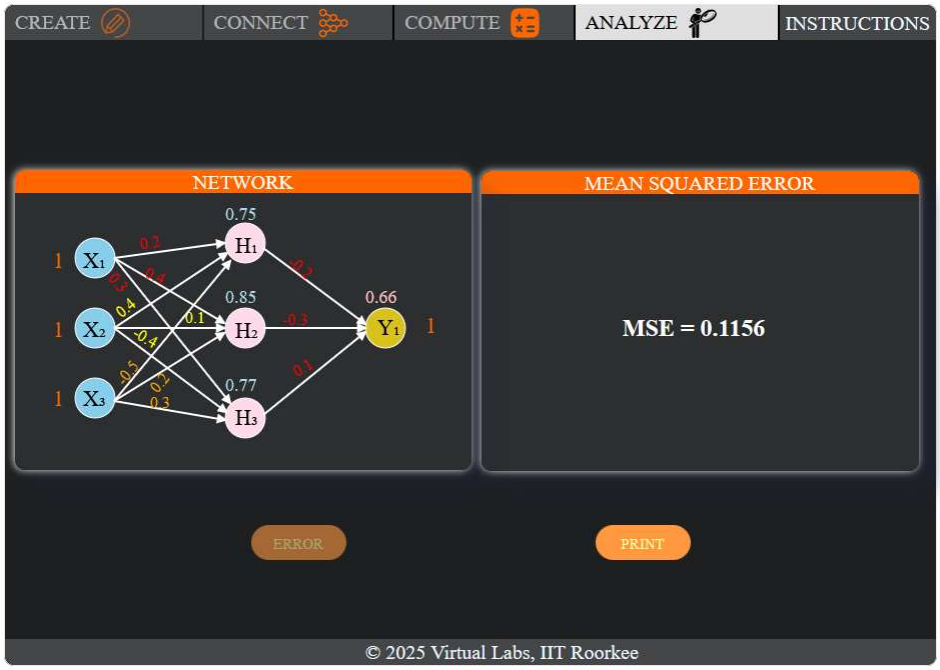


Fig 19: *Final network with computed node outputs and Mean Squared Error (MSE = 0.1156)*

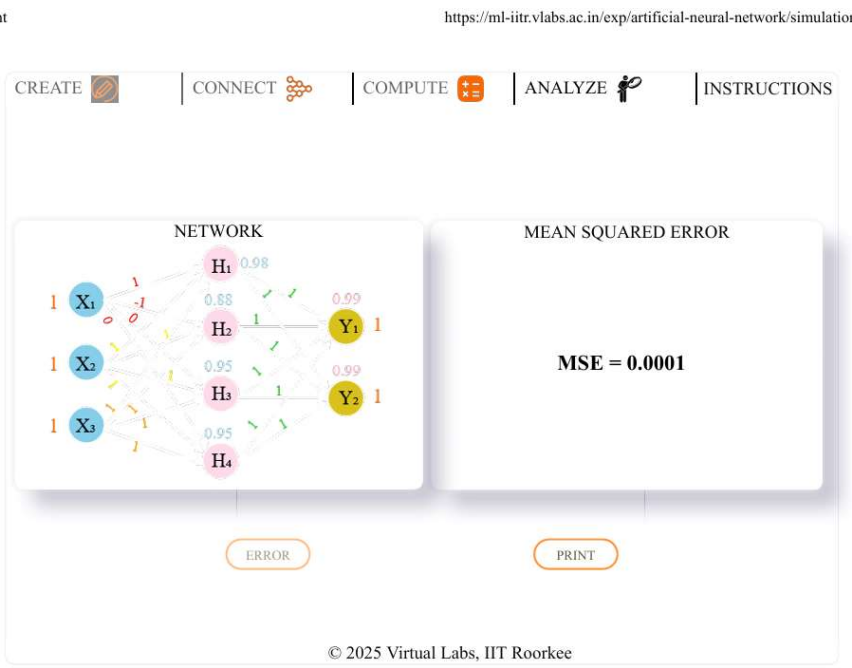


Fig 20: *Printable summary view of the network and Mean Squared Error*

Simulation link: https://ml-iitr.vlabs.ac.in/exp/artificial-neural-network/simulation.html